

chapter-1 Signals, Systems and Signal processing

A signal is defined as any physical quantity that varies with time, space, or any other independent variables.

E.g. speech signal, image signal, $x = \sin \omega t$.

A system is defined as a physical device that performs an operation on a signal.

E.g. a filter used to reduce the noise and interference.

The system is characterized by the type of operation that it performs on the signal.

For example, if the operation is linear, the system is called linear. If the operation is non-linear, the system is said to be nonlinear and so forth. Such operations are usually referred to as signal processing.

Signal analysis describes the field of study whose goal is to collect, understand, and deduce information and intelligence from various signals.

Digital Signal Processing:-

Digital signal processing is the process of analyzing and modifying a signal to optimize or improve its efficiency or performance. It is primarily used to detect errors, and to filter and compress analog signal in transit.

It involves applying various mathematical and computational algorithms to analog and digital signals to produce signals that's of higher quality than the original signal. It facilitates the sharing of a single processor among a no. of signals by time-sharing.

Basic Elements of Digital signal processing:-

Digital signal processing is ~~for~~ provides an alternative method for processing the analog signal as illustrated in fig. below.

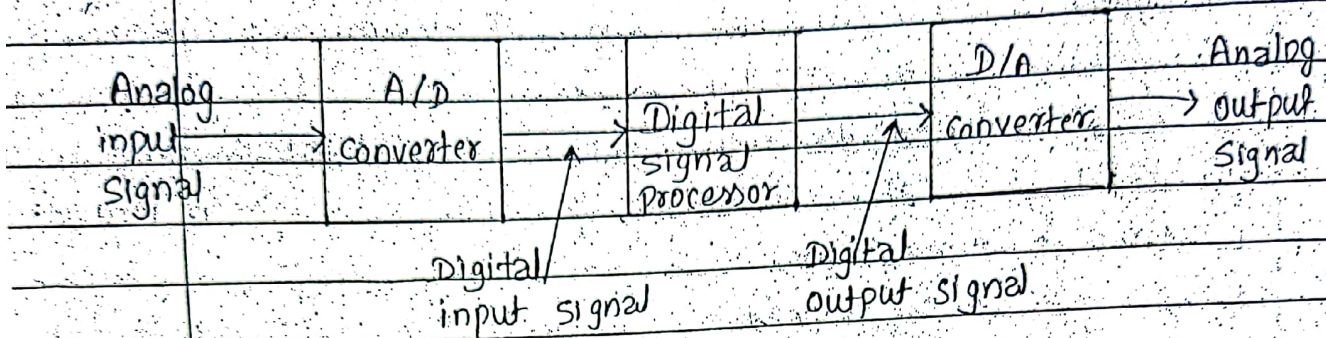


fig. Block diagram of basic elements of Digital signal processing system

Most of the signals generated are analog in nature.

Eg. sound, video, temp^r, pressure, seismic signal etc. If such signals are processed by DSP system, then the signals must be digitized. Hence, input is given through A/D converter and output is obtained through D/A Converter.

The basic elements of DSP system are

- i) A/D Converter
- ii) Digital signal processor
- iii) D/A Converter

i) A/D Converter:-

It converts analog input to digital input, which is appropriate as an input to the digital processor. The A/D Converter determines sampling rate and quantization error in digitizing operation.

ii) Digital Signal processor:

The digital signal processor may be a large programmable digital computer or a small microprocessor programmed to perform the desired operations on the input signal. It performs amplification, attenuation, filtering, spectral analysis, feature extraction etc. operations on digital data.

iii) D/A Converter:

Some of the processed signals are required back in their analog form. For example, sound, image, video are required in analog form. Hence output of digital signal processor is given to D/A converter, which converts digital output of DSP to its analog equivalent.

Advantages of Digital over Analog signal processing:

1. Digital programmable system allows flexibility in reconfiguring the digital signal processing operations simply by changing the program.
2. Better control of accuracy in digital systems compared to analog systems.
3. Digital signals are easily stored on magnetic media (tape or disk) without deterioration or loss of signal. As a consequence, the signals become transportable and can be processed off-line.
4. The digital signal processing method allows for the implementation of more sophisticated signal processing algorithms.
5. Digital implementation of the signal processing system is cheaper than its analog.

6. Digital systems are 'independent of temp.' ageing and other external parameters.
7. Digital circuits are less sensitive to tolerances of component values.
8. Digital system can be cascaded without any loading problems.

Disadvantages of DSP:

1. DSP techniques are limited to signals with relatively low bandwidths i.e. it requires higher bandwidth.
2. Signals having extremely wide bandwidths requires high speed A/D converters and fast digital signal processors, which are difficult to achieve in DSP.
3. The DSP Systems are expensive for small applications.

Applications of DSP:

1. Speech processing:

Speech is 1-dimensional signal. Digital processing of speech is applied to a wide range of speech problems such as speech spectrum analysis, channel vocoders, speech coding etc. DSP is applied to speech enhancement, speech analysis and synthesis, speech recognition.

2. Image processing:

Any 2-dimensional pattern is called an image. Digital processing of images requires 2-dimensional DSP tools such as DFT, FFT algorithms and z-transforms. Processing of electrical signals extracted from images by digital techniques

include image formation and recording, image compression, image reconstruction, image restoration and image enhancement.

3. Radar Signal processing:

Radar stands for "Radio detection and Ranging." improvement in signal processing is possible by digital technology. Development of DSP has led to greater sophistication of radar tracking algorithms.

4. Digital Communication:

Application of DSP in digital communication specially telecommunications comprises of digital transmission using PCM, digital switching using TDM, echo control and digital tape-recorders. DSP in telecommunication systems are found to be cost effective due to ~~small~~ availability of medium and large scale digital ICs.

5. Spectral Analysis:

frequency domain analysis is easily and effectively possible in digital signal processing using FFT algorithms. These algorithm reduce computational complexity and also reduces computational time.

6. Sonar Signal processing:

Sonar stands for "Sound navigation and Ranging". Sonar is used to determine the range velocity and direction of targets that are remote from the observer. DSP can be used to process sonar signals for the purpose of navigation and ranging.

Some more application of DSP

- Control applications
- Biomedical engineering
- Military / Defense applications
- Automotive
- Consumer applications
- Industrial applications
- Entertainment applications
- Telecommunication

A/D and D/A conversion:

Most signals of practical interest such as speech, seismic signals, radar signals, biological signals and various communication signals such as audio and video signals, are analog.

To process analog signal by digital means, it is first necessary to convert them into digital form, i.e. to convert them to a sequence of numbers having finite precision. This procedure is called Analog to digital (A/D) conversion and the corresponding devices are called A/D converters. (ADCs.)

A/D Conversion is a three-step process. This process is illustrated in fig. below.

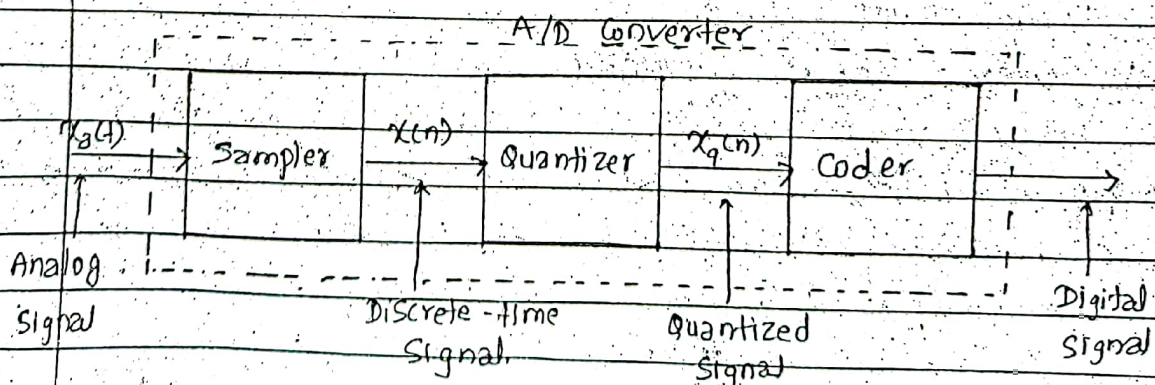


Fig: Basic parts of A/D Converter.

1. Sampling:

This is the conversion of a continuous-time signal into a discrete-time signal obtained by taking "samples" of the continuous-time signal at discrete-time instants. Thus, if $x_a(t)$ is the input to the sampler, the output is $x_a(nT) \equiv x(n)$, where T is called the sampling interval.

2. Quantization:

This is the conversion of a discrete-time continuous-valued signal into a discrete-time, discrete-valued (digital) signal. The value of each signal sample is represented by a value selected from a finite set of possible values. The difference between unquantized sample $x(n)$ and the quantized output $x_q(n)$ is called the quantization error.

3. Coding:

In the coding process, each discrete value $x_q(n)$ is represented by a b -bit binary sequence.

D/A Conversion:-

In many cases of practical interest (e.g. speech processing) it is desirable to convert the processed signal into analog form. The process of converting a digital signal into an analog signal is known as digital to analog (D/A) conversion.

All D/A converts "connect the dots" in a digital signal by performing some kind of interpolation, whose accuracy depends on the quality of the D/A conversion process.

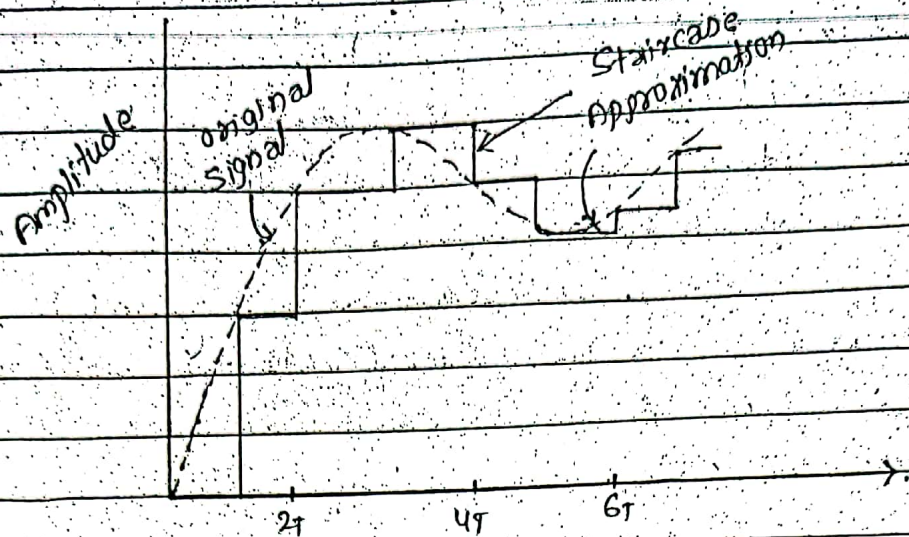
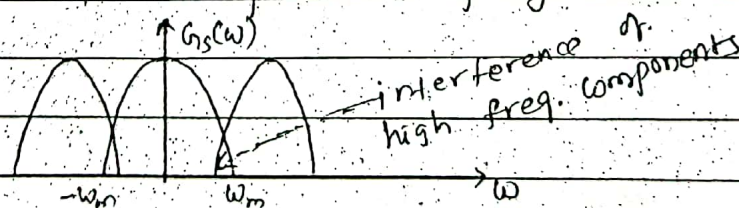


fig: zero-order hold D/A converter

The analog signal can be reconstructed from the samples without any loss of information nor any distortion, provided that the sampling rate is sufficiently high to avoid the problem commonly called Aliasing.

Aliasing is a phenomenon where the high frequency components of the sampled signal interfere with each other because of inadequate sampling.



Aliasing leads to distortion in recovered signal. This is the reason why sampling freq. should be at least twice the bandwidth of the signal.

Sampling of Continuous Signal

Sampling is the process of the conversion of the continuous time signal $x_a(t)$ into discrete-time signal $x(n)$, i.e.

$$x(n) = x_a(nT) \quad -\infty < n < \infty$$

where,

$x(n)$ is the discrete-time signal obtained by "taking samples" of the analog signal $x_a(t)$ every T seconds. This procedure is illustrated in fig. below.

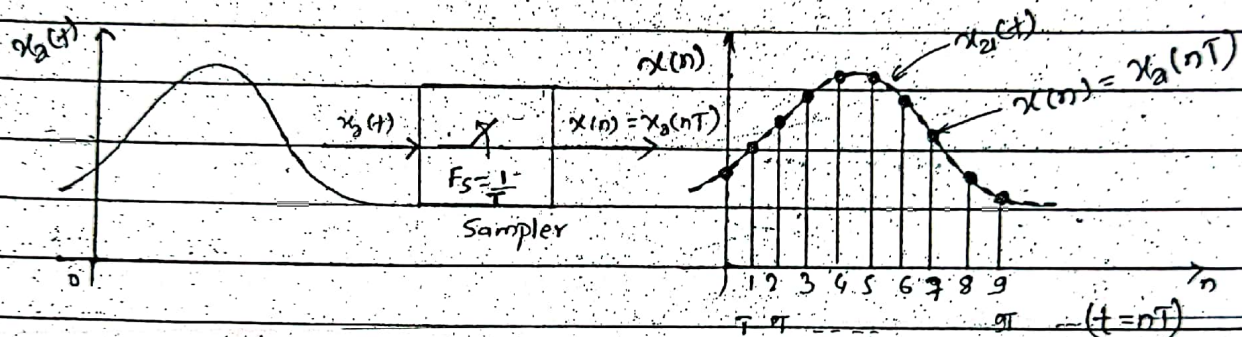


Fig: periodic Sampling of an analog signal.

The time interval T between successive samples is called the sampling period or sampling interval and its reciprocal $\frac{1}{T} = F_s$, is called sampling rate, (samples per second) or sampling freq. (Hz).

Periodic Sampling establishes a relationship between the time variables t and n of continuous-time and discrete-time signals as,

$$t = nT = \frac{n}{F_s}$$

As a result, there exists a relationship between the freq. variable F (or ω) for analog signals and freq. variable

f. (or ω) for discrete time signal.

let us consider an analog sinusoidal signal of the form.

$$x_a(t) = A \cos(2\pi F t + \theta) \quad \text{--- (1)}$$

When sampled periodically at $F_s = \frac{1}{T}$ samples/sec. we get

$$x_a(nT) \equiv x(n) \equiv A \cos(2\pi F nT + \theta) \quad (\because t = nT)$$

$$x(n) = A \cos\left(\frac{2\pi n F}{F_s} + \theta\right) \quad \text{--- (ii)}$$

Comparing eqⁿ (ii) with $x(n) = A \cos(2\pi f n + \theta)$ we get

$$f = \frac{F}{F_s} \quad \text{or,} \quad \omega = \Omega T = \frac{\Omega}{F_s}$$

We know,

$$\text{where, } -\infty < F < \infty \quad \& \quad -\infty < \Omega < \infty$$

Also, we know, that maximum range of discrete time frequencies is

$$-\frac{1}{2} \leq f \leq \frac{1}{2} \quad \text{and} \quad -\pi < \omega < \pi$$

By substituting the value of $f = \frac{F}{F_s}$ and $\omega = \Omega T$, we get

$$-\frac{1}{2} \leq -\frac{F_s}{2} \leq F \leq \frac{F_s}{2} = \frac{1}{2T}$$

$$\text{or, } -\frac{\pi}{T} = -\pi F_s \leq \Omega \leq \pi F_s = \frac{\pi}{T}$$

we observe that the fundamental difference between continuous time and discrete time signals in their range of values of the freq. variables F and f , or Ω and ω .

Thus, periodic sampling of a continuous-time signal implies a mapping of the infinite freq. range for the variable F (or Ω) into a finite freq. range for the variable f (or ω).

Since, the highest freq. in a discrete-time signal is $\omega = \pi$ or $f = \frac{1}{2}$, it follows that, with a sampling rate F_s , the corresponding highest values of F and Ω are

$$F_{\max} = \frac{F_s}{2} = \frac{1}{2T} \quad \text{and} \quad \Omega_{\max} = \pi F_s = \frac{\pi}{T}$$

Sampling Theorem:

It states that, "a continuous time signal can be completely represented in its samples and recovered back if the sampling frequency is $f_s \geq 2f_{\max}$."

$$\text{Nyquist rate} = 2f_{\max}$$

$$\text{Nyquist interval} = \frac{1}{2f_{\max}}$$

$$\text{Resolution, } \Delta = \frac{X_{\max} - X_{\min}}{L - 1}, \quad X_{\max} = A_1 + A_2$$

$$X_{\min} = -X_{\max}$$

$$L = \text{no. of levels.}$$

Continuous-time signals

$$x_a(t) = A \cos(2\pi F t + \theta)$$
$$= A \cos(\omega t + \theta), -\infty < t < \infty$$

$$\omega = 2\pi F \text{ radian}$$

$$\Rightarrow \omega \Rightarrow \frac{\text{radian}}{\text{sec}} = \text{rad/sec}$$

$$\Rightarrow F \Rightarrow \text{Hz}$$

Discrete-time signals

$$x_d[n] = A \cos(2\pi f n + \theta), -\infty < n < \infty$$

$$\text{i.e. } x_d[nT] = x(n)$$

$$\omega = 2\pi f$$

$$\omega \rightarrow \text{radians/sample}$$

$$f \rightarrow \text{cycles/sample}$$

$$\omega = nT, f = \frac{F}{F_s}$$

$$n = \frac{\omega}{T}, F = f \cdot F_s$$

$$-\infty < \omega < \infty$$

$$-\infty < F < \infty$$

$$-\frac{\pi}{T} \leq \omega \leq \frac{\pi}{T}$$

$$-\frac{F_s}{2} \leq F \leq \frac{F_s}{2}$$

8. A digital communication link carries binary coded words representing samples of an input signal x .

$x_0(t) = 3 \cos 600\pi t + 2 \cos 800\pi t$. The link is operated at 10,000 bits/sec and each input sample is quantized into 1024 different voltage levels.

- i) What is the sampling freq. and folding freq.?
- ii) What is the Nyquist rate for the signal $x_0(t)$?
- iii) What are the frequencies in resulting discrete time signal $x(n)$?
- iv) What is the resolution, Δ ?

Soln:- Here, given,

$$x_0(t) = 3 \cos 600\pi t + 2 \cos 800\pi t \quad \text{--- (1)}$$

bit rate = 10,000 bits/sec

no. of quantized level (L) = 1024

let b be the no. of bits/sample.

we know,

$$b = \log_2(L) = \log_2(1024) = 10 \text{ bits/sample}$$

i) we know, Bit rate = Samples/sec $\times$ bits/sample

$$\therefore (F_s) \text{ Samples/sec} = \frac{\text{Bit rate}}{\text{bits/sample}} = \frac{10,000}{10} = 1000 \text{ Hz}$$

Also, we know,

$$\text{Folding freq.} = \frac{F_s}{2} = \frac{1000}{2} = 500 \text{ Hz}$$

∴ sampling freq. (f_s) = 1000 Hz and
folding freq. = 500 Hz.

ii) ⇒ from eqⁿ (i)

$$x_2(t) = 3 \cos 2\pi (300)t + 2 \cos 2\pi (400)t \quad \text{--- (ii)}$$

$$\therefore F_1 = 300 \text{ Hz} \quad \& \quad A_1 = 3$$

$$F_2 = 400 \text{ Hz} \quad \& \quad A_2 = 2$$

$$\text{Nyquist Interval} = \frac{1}{2 f_{\max}} = \frac{1}{2 f_2}$$

$$\therefore f_{\max} = f_2 = 400 \text{ Hz}$$

$$= \frac{1}{2 \times 400}$$

now,

$$= \frac{1}{800} \text{ sec}$$

Nyquist rate is given by

$$\begin{aligned} \text{Nyquist rate} &= 2 f_{\max} \\ &= 2 \times 400 \text{ Hz} \\ &= 800 \text{ Hz} \end{aligned}$$

iii) Replacing t by $\frac{n}{F_s}$ i.e. $t = \frac{n}{F_s}$ in eqⁿ (ii) we get

$$x_2(n) = 3 \cos 2\pi (300) \frac{n}{F_s} + 2 \cos 2\pi (400) \frac{n}{F_s}$$

$$= 3 \cos 2\pi (300) \frac{n}{1000} + 2 \cos 2\pi (400) \frac{n}{1000}$$

$$= 3 \cos 2\pi (0.3) n + 2 \cos 2\pi (0.4) n$$

$$\therefore F_1' = 0.3 \text{ Hz} \quad \& \quad F_2' = 0.4 \text{ Hz}$$

iv)

$$\text{resolution, } \Delta = \frac{x_{\max} - x_{\min}}{L-1} = \frac{A_1 + A_2 - (-(A_1 + A_2))}{1024-1} = \frac{5 - (-5)}{1024-1}$$

$$\therefore \Delta = 0.009775$$

2012-fall
r.b)

- Q. Consider the analog signal, $x_a(t) = 3 \cos 100\pi t$,
- Determine the min. sampling rate required to avoid aliasing.
 - Suppose that the signal is sampled at the rate $F_s = 200$ Hz, what is the discrete-time signal obtained after sampling?
 - Suppose that the signal is sampled at the rate $F_s = 75$ Hz, what is the discrete-time signal obtained after sampling?
 - What is the freq. $0 < F < F_s$ of a sinusoid that yields samples identical to those obtained in part (ii)?

Solⁿ:- i) $\Rightarrow$ Here, $x_a(t) = 3 \cos 100\pi t$
 $x_a(t) = 3 \cos 2\pi (50)t$ — (1)

$\therefore F = 50$ Hz.

$\therefore$ The freq. of the analog signal is $F = 50$ Hz.

According to Nyquist rate,

i.e. Nyquist rate $(F_s) \geq 2 F_{\max} = 2 \times 50 \text{ Hz} = 100 \text{ Hz}$

Hence, the min. sampling rate required to avoid aliasing is

$F_s = 100$ Hz.

- ii) $\Rightarrow$ If signal is sampled at $F_s = 200$ Hz, the discrete-time signal is (i.e. put $t = \frac{n}{F_s}$ in eqⁿ (1))

$x(n) = 3 \cos 2\pi 50 \left(\frac{n}{200} \right)$

$\therefore x(n) = 3 \cos \left(\frac{\pi}{2} \right) n$

- iii) If signal is sampled at $F_s = 75$ Hz, then

$x(n) = 3 \cos 2\pi 50 \left(\frac{n}{75} \right) = 3 \cos \left(\frac{4\pi}{3} \right) n$

or, $x(n) = 3 \cos \left(2\pi - \frac{2\pi}{3} \right) n = 3 \cos \frac{2\pi}{3} n$

iv) $\Rightarrow$ for the sampling rate of $F_s = 75 \text{ Hz}$ we have

$$F = f F_s = 75 f$$

The freq. of the sinusoid in part (iii) is

$$f = \frac{1}{2}$$

$$\text{Hence } F = 75 \times \frac{1}{2} = 25 \text{ Hz}$$

clearly, the sinusoid signal

$$\begin{aligned} y_2(t) &= 3 \cos 2\pi F t \\ &= 3 \cos 50\pi t \end{aligned}$$

Sampled at $F_s = 75$ samples/sec yields identical samples.

Hence $F = 50 \text{ Hz}$ is an alias of $F = 25 \text{ Hz}$ for the sampling rate $F_s = 75 \text{ Hz}$.

2017-Spring

Q. for a digital system with a max. audio input frequency of 6 kHz , determine the min. sample rate and the alias frequency produced if a 7 kHz audio signal were allowed to enter the analog to digital converter.

Solⁿ: Here, given, let $f_{\max}$ be the freq. of audio.

$$\therefore f_{\max} = 6 \text{ kHz}$$

$$\text{Min Sample rate} \geq 2 f_{\max} = 2 f_{\max} = 2 \times 6 \text{ kHz} = 12 \text{ kHz}$$

$$\text{Sampling freq. } (f_s) = 7 \text{ kHz}$$

Q. Consider the signal $x_2(t) = 10 \cos 2\pi(1000)t + 5 \cos 2\pi(5000)t$ is to be sampled.

- i) determine the nyquist rate for this signal
- ii) If the signal is sampled at 4 kHz , will the signal be recovered from its samples?

Soln: Here, $x_2(t) = 10 \cos 2\pi(1000)t + 5 \cos 2\pi(5000)t$

Clearly, the signal $x_2(t)$ contains two cosine waves

$$F_1 = 1000 \text{ Hz} \quad \& \quad A_1 = 10$$

$$F_2 = 5000 \text{ Hz} \quad \& \quad A_2 = 5$$

- i). Since, the highest freq. in $x_2(t)$ is $F_{\max} = F_2 = 5000 \text{ Hz}$

According to nyquist criteria,

$$f_{\text{sam}} \geq 2 f_{\max} \Rightarrow f_{\text{sample}} = 2 \times 5000 = 10000 \text{ Hz}$$

- ii) : Since, the signal is sampled at $f_s = 4 \text{ kHz} = 4000 \text{ Hz}$
When $x_2(t)$ is sampled, we obtain $x(n)$, by $t \rightarrow \frac{n}{f_s}$

$$x(n) = 10 \cos 2\pi (1000) \frac{n}{4000} + 5 \cos 2\pi (5000) \frac{n}{4000}$$

$$= 10 \cos 2\pi \left(\frac{1}{4}\right) n + 5 \cos 2\pi \left(\frac{5}{4}\right) n$$

$$= 10 \cos 2\pi \left(\frac{1}{4}\right) n + 5 \cos 2\pi \left(1 + \frac{1}{4}\right) n$$

$$= 10 \cos 2\pi \left(\frac{1}{4}\right) n + 5 \cos 2\pi \left(\frac{1}{4}\right) n$$

$$\therefore x(n) = 15 \cos 2\pi \left(\frac{1}{4}\right) n$$

which is sampled signal.

When this signal is reconstructed, then

$$x_r(t) = 15 \cos 2\pi (1000) t \quad \because n = t F_s = 4000t$$

Here, we observe that the reconstructed signal contains only one freq. of 1000 Hz and amplitude is 15.

Thus, the effect of $F_2 = 5000$ Hz is completely lost and amplitude of F_1 is increased. This shows that with the sampling rate 4000 Hz, the signal is not recovered from its samples. This is because the sampling theorem is not satisfied.

The Nyquist rate as calculated in (i) is 10 kHz.

$\therefore$ min. sampling freq. should be 10 kHz to avoid aliasing.

2017-18
Q.

The discrete time signal $x(n) = 6.5 \cos(0.1\pi n)$ is quantized with a resolution, $\Delta = 0.01$. How many bits are required in the A/D converter? If the

max. freq. that can be reconstructed from above signal is 500 Hz, determine the reconstructed continuous time signal.

Soln: Here given, $x(n) = 6.5 \cos(0.1 \pi n)$
 $= 6.5 \cos 2\pi \left(\frac{1}{20}\right) n$

$\therefore$ freq. (F_1) = $\frac{1}{20}$ Hz, & $A_1 = 6.5$

resolution, $\Delta = 0.01$, & max. freq. ($f_{\max}$) = 500 Hz.

$\therefore$ Sampling freq. (f_s) = $2 f_{\max} = 2 \times 500$ Hz = 1000 Hz.

We know, $\Delta = \frac{x_{\max} - x_{\min}}{L-1} = \frac{A_1 - (-A_1)}{L-1}$

or, $0.01 = \frac{6.5 + 6.5}{L-1} \Rightarrow L = 1301$

now, no. of bits/sample = $\log_2(L) = \log_2(1301) = 10.345$

Let $x_2(t)$ is reconstructed continuous time signal

So,

$x_2(t) = 6.5 \cos 2\pi(1000 \times 0.05)t$ $n = tF_s = 1000t$

$\therefore x_2(t) = 6.5 \cos 2\pi(50)t$

2015-2016
 1. b) Q. A digital communication link binary-coded words representing samples of an input signal

$x_2(t) = 3 \cos 5000\pi t + 2 \sin 6000\pi t$

The link is operated at 18,000 bits/sec and each input sample is quantized into 512 different voltage levels.

- i) what is the discrete-time signal obtained after sampling
 ii) what is resolution, Δ ?

Soln: we know, $b = \log_2 L = \log_2(512) = 9$ bits/sample

we know,

$$F_s (\text{samples/sec}) = \frac{\text{bit rate}}{\text{bits/sample}} = \frac{18000}{9} = 2000 \text{ samples/sec}$$

$$\therefore \text{Sampling freq. } (F_s) = 2000 \text{ samples/sec}$$

Here,

$$x_2(t) = 3 \cos 5000\pi t + 2 \sin 6000\pi t$$

$$\therefore x_2(t) = 3 \cos 2\pi(2500)t + 2 \sin 2\pi(3000)t$$

which has, $F_1 = 2500 \text{ Hz}$ & $A_1 = 3$

$F_2 = 3000 \text{ Hz}$ & $A_2 = 2$

After sampling, i.e. $t \rightarrow \frac{n}{F_s}$ or, $f = \frac{F}{F_s}$

$$x(n) = 3 \cos 2\pi \left(\frac{2500}{2000} \right) n + 2 \sin 2\pi \left(\frac{3000}{2000} \right) n$$

$$\therefore x(n) = 3 \cos 2\pi(1.25)n + 2 \sin 2\pi(1.5)n$$

which is the required discrete-time signal obtained after sampling.

ii) $\Rightarrow$ we know, resolution (Δ) = $\frac{X_{\max} - X_{\min}}{L-1}$

$$= \frac{5 - (-5)}{512-1}$$

$$\therefore \Delta = 0.01956$$

$$X_{\max} = A_1 + A_2 = 5$$

$$X_{\min} = -X_{\max} = -5$$